Practice Set A

Class 12 Mathematics, Chapter 2: Inverse Trigonometric Functions

Maximum marks: **43** **20** questions

Name _____ Date _____ Score _____ / 43

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 MCQ, PYQ-style — 1 mark

The principal value of $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ is:

(A) $-\frac{\pi}{3}$ **(B)** $\frac{\pi}{3}$ **(C)** $\frac{2\pi}{3}$ **(D)** $\frac{5\pi}{6}$

2 MCQ, PYQ-style — 1 mark

The value of $\cos^{-1}\left(\sin\frac{4\pi}{3}\right)$ is:

(A) $\frac{4\pi}{3}$ **(B)** $-\frac{\pi}{6}$ **(C)** $\frac{5\pi}{6}$ **(D)** $\frac{\pi}{6}$

3 MCQ, CBSE SQP 2025-26 — 1 mark

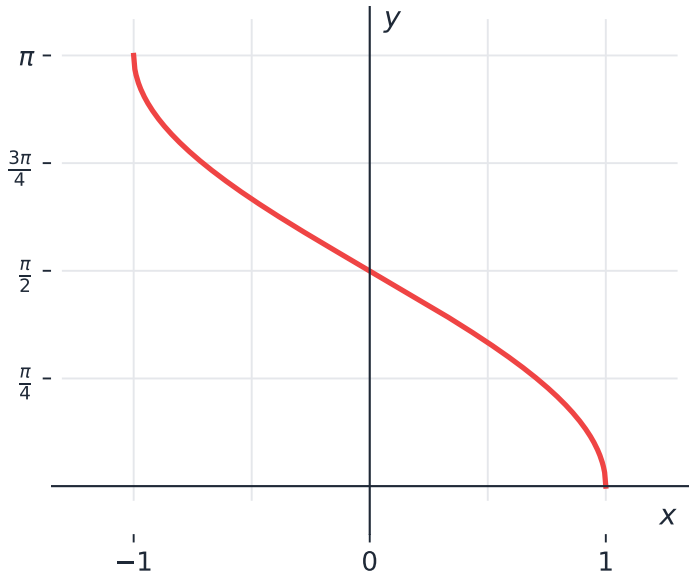
The inverse trigonometric function whose domain is $\left[-\frac{1}{3},\frac{1}{3}\right]$ is:

(A) $\cos^{-1}x$ **(B)** $\cos^{-1}\left(\frac{x}{3}\right)$ **(C)** $\cos^{-1}(3x)$ **(D)** $3\cos^{-1}x$

Q No.	Question	Marks

4 MCQ, CBSE SQP 2024-25 — 1 mark

The graph drawn below depicts:



(A) $y = \sin^{-1} x$ **(B)** $y = \cos^{-1} x$

(C) $y = \operatorname{cosec}^{-1} x$ **(D)** $y = \cot^{-1} x$

5 MCQ, PYQ-style — 1 mark

The principal value of $\cos^{-1}\left(-\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$ is:

(A) 0 **(B)** $\frac{5\pi}{6}$ **(C)** π **(D)** $\frac{4\pi}{3}$

6 Assertion-Reason, CBSE SQP 2025-26 — 1 mark

Assertion (A): The value of the expression $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) + \tan^{-1} 1 - \sec^{-1}\left(\sqrt{2}\right)$ is $\frac{\pi}{4}$.

Reason (R): The principal value branch of $\sin^{-1} x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and that of $\sec^{-1} x$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

7 Short answer, CBSE 2023 — 2 marks

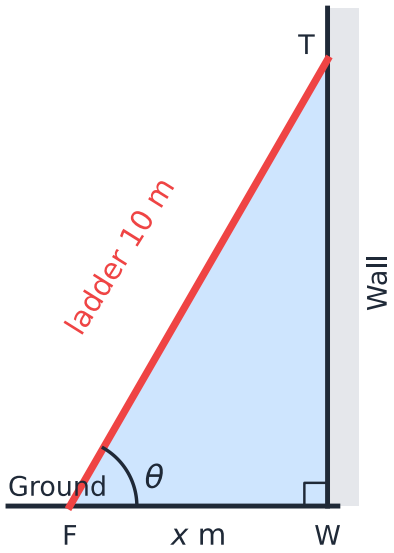
Evaluate: $\sin^{-1}\left(\sin\frac{3\pi}{4}\right) + \cos^{-1}(\cos\pi) + \tan^{-1}(1)$.

Q No.	Question	Marks
Level 2		**16 marks**
8	Short answer, CBSE 2023 Evaluate: $3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)+2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)+\cos^{-1}(0)$.	2 marks
9	Short answer, CBSE 2025 Find the domain of $f(x)=\sin^{-1}\left(-x^2\right)$.	2 marks
10	Short answer, CBSE 2025 Find the domain of $\sec^{-1}(2x+1)$.	2 marks
11	Short answer, CBSE 2024 (Comptt.) Evaluate: $\sin^{-1}\left(\sin\frac{13\pi}{6}\right)+\cos^{-1}\left(\cos\frac{\pi}{3}\right)+\tan^{-1}\left(\sqrt{3}\right)$.	2 marks
12	Short answer, CBSE SQP 2025-26 Evaluate: $\tan\left(\tan^{-1}(-1)+\frac{\pi}{3}\right)$.	2 marks
13	Short answer, CBSE 2024 Find the value of k if $\sin^{-1}\left[k\tan\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]=\frac{\pi}{3}$.	3 marks
14	Short answer, CBSE 2024 Find the value of $\sin^2\left(\cos^{-1}\frac{3}{5}\right)+\tan^2\left(\sec^{-1}3\right)$.	3 marks
Level 3		**19 marks**
15	Long answer, CBSE 2023 Draw the graph of $f(x)=\sin^{-1}x$, $x\in\left[-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right]$. Also, write the range of $f(x)$.	3 marks
16	Long answer, CBSE 2024 Find the value of $\sin^{-1}\left(-\frac{1}{2}\right)+\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)+\cot^{-1}\left(\tan\frac{4\pi}{3}\right)$.	3 marks
17	Long answer, CBSE 2024 Find the domain of $f(x)=\cos^{-1}\left(1-x^2\right)$. Also, find its range.	3 marks
18	Long answer, CBSE 2024 Simplify: $\cos^{-1}x+\cos^{-1}\left[\frac{x}{2}+\frac{\sqrt{3-3x^2}}{2}\right]$, $\frac{1}{2}\le x\le 1$.	3 marks
19	Long answer, CBSE 2024 Express $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right)$, $-\frac{\pi}{2}<x<\frac{\pi}{2}$, in the simplest form.	3 marks

Q No.	Question	Marks

20 Case study, Board-pattern — 4 marks

Case Study: A ladder 10 m long rests against a vertical wall. Its foot F is on level ground, x m from the foot W of the wall, and its top T touches the wall. The angle θ that the ladder makes with the ground is given by $\theta = \cos^{-1}\left(\frac{x}{10}\right)$.



(i) Find θ when $x = 5$ m. — 1 mark

(ii) Write the domain of the function $f(x) = \cos^{-1}\left(\frac{x}{10}\right)$ and the range of its principal value branch. — 1 mark

(iii) The foot of the ladder slips away from the wall until $\theta = \frac{\pi}{6}$. Find the new distance x of the foot from the wall and the height WT at which the top now touches the wall. — 2 marks

Answer key

Chapter 2: Inverse Trigonometric Functions, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(C) $\dfrac{2\pi}{3}$
2	(C) $\dfrac{5\pi}{6}$
3	(C) $\cos^{-1}(3x)$
4	(B) $y = \cos^{-1} x$
5	(C) π
6	(D) Assertion (A) is false, but Reason (R) is true.
7	$\dfrac{3\pi}{2}$

Level 2

8	$\dfrac{19\pi}{12}$
9	Domain $= [-1, 1]$
10	Domain $= (-\infty, -1] \cup [0, \infty)$
11	$\dfrac{5\pi}{6}$
12	$2 - \sqrt{3}$
13	$k = \dfrac{1}{2}$
14	$\dfrac{216}{25}$

Level 3

15	Graph: the increasing arc of $y = \sin^{-1} x$ from $\left(-\dfrac{1}{\sqrt{2}}, -\dfrac{\pi}{4}\right)$ through $(0, 0)$ to $\left(\dfrac{1}{\sqrt{2}}, \dfrac{\pi}{4}\right)$; range $= \left[-\dfrac{\pi}{4}, \dfrac{\pi}{4}\right]$.
16	$\dfrac{5\pi}{6}$
17	Domain $= \left[-\sqrt{2}, \sqrt{2}\right]$; range $= [0, \pi]$
18	$\dfrac{\pi}{3}$
19	$\dfrac{\pi}{4} + \dfrac{x}{2}$
20	(i) $\theta = \dfrac{\pi}{3}$ (ii) Domain $[-10, 10]$; range $[0, \pi]$ (iii) $x = 5\sqrt{3} \approx 8.66$ m; height $= 5$ m